

Feature Test Cases

Priority Legend:

■ High — core functionality

■ Medium — important but not blocking

■ Low — nice-to-have /
cosmetic

TC-01

■ Persistent Cross-Session Memory

■ High

Description

The AI remembers summaries of past conversations and injects them into new chats automatically.

Pre-conditions

- App is open at the Vercel URL
- At least one LLM API key is configured

Test Steps

1. Start a chat and send: "My name is Alex and I am studying for AZ-900"
2. Send 3 more messages on any topic
3. Close the browser tab completely
4. Reopen the app in a new tab
5. Send: "What do you know about me?"

Expected Result

The AI responds with the user's name (Alex) and study goal (AZ-900) without being told again.

Result: ■ PASS

■ FAIL

Notes / Defect ID: _____

TC-02

■ Session History Panel

■ High

Description

Browse all past conversations by auto-generated title and resume any session with full context.

Pre-conditions

- At least one previous conversation exists
- Backend is online

Test Steps

1. Have a conversation — send at least 3 messages
2. Click the ■ (clock) button in the chat header
3. Verify current session appears under "Current session"
4. Click "+ New Chat" to start a fresh session
5. Re-open the ■ panel
6. Click the previous session from the "Past sessions" list

Expected Result

The full previous conversation reloads in the chat window. The session ID updates.

Result: ■ PASS

■ FAIL

Notes / Defect ID: _____

TC-03**■ Learning Progress Dashboard****■ Medium**

Description Shows grade (A–F), quiz accuracy bars, streak, topics covered, mode breakdown, and exam history.

Pre-conditions • At least 3 quiz questions have been answered

Test Steps

1. Switch to Quiz mode using the mode tabs at the top of the chat
2. Answer 4 questions — answer 3 correctly and 1 incorrectly
3. Open the Learning Progress panel in the left sidebar
4. Observe the grade badge, accuracy bar, and mode breakdown

Expected Result Grade shows B or higher ($\geq 75\%$), quiz accuracy bar shows ~75%, 'Quiz' appears in mode breakdown.

Result: ■ PASS ■ FAIL Notes / Defect ID: _____

TC-04**■ Exam Mode****■ High**

Description Structured graded exam with 5 varied question types, automatic scoring, grade, and improvement suggestions.

Pre-conditions

- Exam mode is available in mode tabs
- LLM API key is configured

Test Steps

1. Click the Exam tab (■) in the mode selector
2. Send: "Give me an exam on Python basics"
3. Answer all 5 questions in the exam
4. Type SUBMIT to trigger grading

Expected Result The AI returns a score (e.g. 4/5), a letter grade (e.g. B), and a bullet list of topics to improve.

Result: ■ PASS ■ FAIL Notes / Defect ID: _____

TC-05**■ Certification Prep (■ Panel)****■ High****Description**

20 certifications with mock exams in the exact vendor format, coaching mode, and study plans.

Pre-conditions

- LLM API key is configured

Test Steps

1. Click the ■ (trophy) button in the chat header
2. Type 'AZ-900' in the search box
3. Select the AZ-900 certification card
4. Review the domain breakdown and stats
5. Click 'Start Mock Exam'

Expected Result

Chat starts an AZ-900 mock exam with Microsoft-style questions covering Cloud Concepts, Azure Services, and Pricing domains.

Result: ■ PASS

■ FAIL

Notes / Defect ID: _____

TC-06**■ Coaching Mode****■ Medium****Description**

Interactive one-question-at-a-time practice for any subject or certification with instant feedback.

Pre-conditions

- LLM API key is configured

Test Steps

1. Click the ■ button and select any certification (e.g. AWS SAA-C03)
2. Click 'Coach Me'
3. Answer the first question
4. Observe feedback and wait for next question

Expected Result

The AI gives immediate feedback on the answer, explains why it is correct or incorrect, then asks the next question.

Result: ■ PASS

■ FAIL

Notes / Defect ID: _____

TC-07**■ Web Search Integration****■ Medium**

Description	Detects search intent, queries DuckDuckGo, and augments the AI response with live results.
Pre-conditions	• Internet connection is available on the backend • LLM API key is configured
Test Steps	1. In the chat input, type: "Search for the latest Python 3.13 release notes" 2. Press Send 3. Wait for the search indicator to appear
Expected Result	A '■ Searching the web for...' message appears, followed by a response referencing current Python 3.13 information.

Result: ■ PASS ■ FAIL Notes / Defect ID: _____

TC-08**■ User Profile Personalisation****■ Medium**

Description	User sets name, background, and goals; the AI adapts tone and depth to the user's level in every reply.
Pre-conditions	• My Profile panel is visible in the sidebar
Test Steps	1. Open My Profile in the sidebar and click Edit 2. Enter: Name = 'Jordan', Background = 'Software developer', Goal = 'Learn cloud architecture' 3. Save the profile 4. In the chat, send: "Explain containers"
Expected Result	The response uses developer-appropriate language and relates containers to cloud architecture without the user restating their background.

Result: ■ PASS ■ FAIL Notes / Defect ID: _____

TC-09**■ URL / Media Ingestion****■ Medium**

Description	Paste any URL into chat; the app fetches the content, adds it to the knowledge base, and answers questions about it.
Pre-conditions	• Backend is online • LLM API key is configured • Valid internet-accessible URL is available
Test Steps	1. In the chat, type: "Load https://en.wikipedia.org/wiki/Machine_learning and summarise it" 2. Press Send 3. Wait for the confirmation message
Expected Result	'■ Loaded [URL] (N chunks)' confirmation appears, followed by a summary drawn from the Wikipedia article content.

Result: ■ PASS ■ FAIL Notes / Defect ID: _____

TC-10**■ Document Upload & RAG****■ High**

Description	Upload PDF, DOCX, MD, or TXT files; the AI uses document chunks to answer questions accurately.
Pre-conditions	• A test PDF or DOCX file is available locally • LLM API key is configured
Test Steps	1. Click the File tab in the Upload panel on the left 2. Drag and drop a PDF file onto the upload area 3. Wait for the success status (green tick) 4. In the chat, ask a specific question from the document content
Expected Result	The AI answers using exact details from the uploaded document and cites the filename as a source.

Result: ■ PASS ■ FAIL Notes / Defect ID: _____

TC-11

■ Multi-Provider AI Switching

■ High

Description	Switch between Claude, Gemini, and OpenAI mid-conversation using the Provider panel or voice commands.
Pre-conditions	• At least two API keys are configured (e.g. CLAUDE_API_KEY + GEMINI_API_KEY)
Test Steps	1. Note the active provider shown in the AI Provider panel (e.g. Claude — Agent 1) 2. In the chat, type: "Hey Agent 2" 3. Observe the confirmation message 4. Ask any question
Expected Result	The app confirms 'Switched to Gemini (Agent 2)' and the next AI response comes from the Gemini model.

Result: ■ PASS ■ FAIL Notes / Defect ID: _____

TC-12

■ Flashcard Mode

■ Low

Description	Generates 5 Q&A; flashcard pairs from uploaded content or general knowledge on any topic.
Pre-conditions	• LLM API key is configured
Test Steps	1. Switch to Flashcards mode using the mode tab 2. Send: "Generate flashcards on the OSI model"
Expected Result	The AI returns exactly 5 numbered flashcard pairs with a clear question and concise answer for each of the 7 OSI layers.

Result: ■ PASS ■ FAIL Notes / Defect ID: _____

TC-13

■■ Streaming Responses & Stop Button

■ Medium

Description	Responses stream token-by-token in real time; the user can stop generation mid-stream.
Pre-conditions	• LLM API key is configured
Test Steps	1. Send a long-answer prompt: "Explain the entire history of computing in detail" 2. While the response is streaming, click the ■■ Stop button
Expected Result	The response stops immediately at the current token. No further text is appended. The input box re-enables.

Result: ■ PASS ■ FAIL Notes / Defect ID: _____

TC-14**■ Dark / Light Theme Toggle****■ Low****Description**

The entire UI switches between dark and light themes; preference persists across page reloads.

Pre-conditions

- App is loaded in the browser

Test Steps

1. Note the current theme (light or dark)
2. Click the ■ / ■■ toggle button in the top-right of the header
3. Reload the page (F5)

Expected Result

The theme switches instantly. After reload, the same theme is still active (persisted in localStorage).

Result: ■ PASS

■ FAIL

Notes / Defect ID: _____

TC-15**■■ Error Log & Diagnostics Panel****■ Low****Description**

All errors (upload failures, API errors, network issues) are captured and shown in the Error Log panel.

Pre-conditions

- App is loaded in the browser

Test Steps

1. Temporarily disconnect from the internet (or stop the backend)
2. Attempt to send a chat message
3. Reconnect to the internet
4. Open the Error Log panel in the sidebar

Expected Result

The failed request appears in the Error Log with a timestamp, source (Chat), and error message. A red badge count shows in the header.

Result: ■ PASS

■ FAIL

Notes / Defect ID: _____

Tester**Name:** _____**Date:** _____**Sign-off:** _____